# Heat Equation with Finite Element Analysis

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**Math 675 Spring 2026:**
**Project 2 Notes**
**Due Thursday, May 14th, 2026**

## Progress Checklist

- **Introduction:** (Problem, motivation, and context)
  Completed
- **Background:** (Definitions and setup)
  Completed
- **Model Development:** (Derivation and assumptions)
  Almost Done, needs images and more details
- **Analysis:** (Mathematical results and explanations)
  In Progress, needs images and more details
- **Discussion:** (Interpretation, limitations, and data considerations)

- **Real World Data Comparison:** (Comparison to actual dataset)
  Completed, squeezed as much info as i could from the code and dataset, but needs images and more details
- **Analysis:** (Mathematical results and explanations)
  In Progress, needs images and more details
- **Discussion:** (Interpretation, limitations, and data considerations)

- **Deeper Dive?** (Extending ideas and exploring more complex models)

- **Conclusion:** (Summary and extensions)

- **Reflection:** (Audio Feeback changes and RECENT EMAIL NOTES

- **AI Usage Citations:** (Clearly cite AI and explain how it was used)

- **References:** (Literature and sources)
  Completed, but should filter out the unnecessary ones
- **Include more images than poster:**
- **Final Formatting Tweaks:**
- **Final Page Count excluding Images:**
- **Summary of work on Presentation:**

## Introduction

If someone were to give you along rod that is heated at one end and cooled at the other, and that person told you to provide the temperature at a particular point along the rod at a particular time, you would not be automaically able to calculate that value without some external help or algorithms. We would want to solve a partial differential equation that describes how heat diffuses through the rod, which is called the heat equation, to understand how the temperature changes over time and space.

But because this is a difficult equation to solve analytically and even more complicated to obtain a nice closed form solution, we can use an iterative method to approximate the temperature distribution along the rod at discrete points in space and time. This method allows us to solve this equation, we can use the finite element method (FEM) to discretize the spatial domain and then apply a time-stepping scheme to evolve the solution over time.

FEM is quite similar to a finite difference method, but as we deal with more complex geometries and higher dimensions, the finite element method becomes more advantageous due to its flexibility in handling irregular meshes and complex boundary conditions.

## Background

### The Governing Heat Equation

For a one-dimensional rod, the heat conduction is governed by:

$$\rho c_p \frac{\partial T}{\partial t} = k \frac{\partial^2 T}{\partial x^2}$$

- $\frac{\partial T}{\partial t}$ = the rate of change of temperature with respect to time ($^\circ C/s$)
- $\frac{\partial^2 T}{\partial x^2}$ = sensitivity of temperature changes at a specific point from its neighbors ($^\circ C/m^2$)

- $t$ = Time (s), • $x$ = Position along the rod ($m$), • $\alpha = \frac{k}{\rho c_p}$ is the thermal diffusivity ($m^2/s$)
- $\rho$ = Density ($kg/m^3$) • $k$ = Thermal Conductivity ($W/m \cdot^\circ C$), • $c_p$ = Specific Heat ($J/kg \cdot^\circ C$)

If we substitute in these units into the heat equation, we can verify that the units are consistent:

$$\left(\frac{kg}{m^3} \cdot \frac{J}{kg \cdot^\circ C} \cdot \frac{^\circ C}{s}\right) = \left(\frac{W}{m \cdot^\circ C} \cdot \frac{^\circ C}{m^2}\right)$$

$$\left(\frac{J}{m^3 \cdot^\circ C \cdot s}\right) = \left(\frac{J}{s \cdot m^3 \cdot^\circ C}\right)$$

This confirms that the units on both sides of the equation are consistent, validating the physical correctness of the heat equation in terms of dimensional analysis. This new unit is called the Volumetric Heat Capacity, which represents the amount of heat required to raise the temperature of a unit volume of the material by one degree Celsius. It is a crucial parameter in understanding how materials respond to thermal energy and is commonly used in heat transfer calculations and thermal analysis.

**How do we build the heat equation from scratch?**

| **Phase** | **Mathematical Logic** | **Summary of each Step** |
|---|---|---|
| **1. Net Energy Flow** | $\frac{dE}{dt} = (E_{in} - E_{out}) - E_{loss}$ | Conservation of energy: Change equals net conduction minus environmental loss. |
| **2. Storage** | $m = \rho V$<br>$E = mc_pT$<br>$\frac{dE}{dt} = \rho V c_p \frac{dT}{dt}$ | Mass is density times volume. Energy storage links mass and specific heat to temperature change. Differentiating now gives us a new relationship |
| **3. Conduction** | $E_{in} - E_{out} = -V\frac{\partial}{\partial x}(E_{flux})$ | Net heat movement is the spatial change of energy flux over the slice volume $V$. |
| *Fourier's Law* | $E_{flux} = -k\frac{\partial T}{\partial x}$ | The energy flux, representing flow intensity, depends on the material's thermal conductivity $k$. |
| *Net Result* | $E_{in} - E_{out} = Vk\frac{\partial^2 T}{\partial x^2}$ | This result gives a second derivative (curvature), which forces heat to spread away from hot peaks |
| **4. Cooling** | $E_{loss} = V\nu(T - T_\infty)$ | Energy loss to air ($T_\infty = 20.8$) using the effective cooling coefficient $\nu$. |
| **5. Substitution** | $\rho V c_p\frac{dT}{dt} = Vk\frac{\partial^2 T}{\partial x^2} - V\nu(T - T_\infty)$ | Substitute the movement and loss terms back into the original balance equation. |
| **6. Final PDE** | $\frac{dT}{dt} = \alpha\frac{\partial^2 T}{\partial x^2} - \nu(T - T_\infty)$ | Divide by $\rho V c_p$ to solve for $\frac{dT}{dt}$ using diffusivity $\alpha = \frac{k}{\rho c_p}$, which produces our Heat Equation!! |

# Model Development

## Boundary and Initial Conditions

The model uses the Dirichlet boundary conditions. Meaning the temperature never changes at the boundaries yet allows every other value to vary. The specific conditions are:

$$T(0,t) = 100°\text{F} \quad \text{(Left Boundary)}$$
$$T(L,t) = 20°\text{F} \quad \text{(Right Boundary)}$$
$$T(x,0) = T(x) \quad \text{(Initial Temperature Gradient)}$$

## Finite Element Analysis

The rod is divided into $n$ equally spaced elements of length $h = L/n$. This discretization allows us to approximate the continuous temperature distribution along the rod using a finite number of points. But this results in a system of equations that can be solved using numerical methods.

**1: Spatial Discretization (FEM)**

For each element along the rod, the local stiffness or local connectivity matrix, $\mathbf{K}^{(e)}$, how each element connects to its neighbors and transfers heat, is defined as:

$$\mathbf{K}^{(e)} = \frac{k}{h}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

**2. Transient System (Backward Euler):**

To ensure unconditional stability, an implicit scheme is used:

$$(\mathbf{I} - \alpha\Delta t\mathbf{K}')\mathbf{T}^{n+1} = \mathbf{T}^n + \mathbf{b}_{bc}$$

Let's just quickly derive this Backward Euler scheme for our transient heat equation. Starting with the heat equation:

$$\frac{\partial T}{\partial t} = \alpha\frac{\partial^2 T}{\partial x^2}$$

We can discretize the time derivative using a backward difference approximation and our spatial second derivative using the finite element method from above rather than a finite difference method. The backward difference approximation for the time derivative is:

$$\frac{T^{n+1} - T^n}{\Delta t}\mathbf{I} = \alpha\mathbf{K}^{(e)}$$

Rearranging this gives us to solve for the temperature at the next time step given the current temperature distribution:

$$T^{n+1}\mathbf{I} = T^n\mathbf{I} + \alpha\Delta t\mathbf{K}^{(e)}$$

Now, we can express the spatial second derivative using the finite element method, which leads to a system of equations that can be represented in matrix form as:

$$(\mathbf{I} - \alpha \Delta t \mathbf{K}')\mathbf{T}^{n+1} = \mathbf{T}^n$$

$$\frac{dT}{dt} = \alpha \frac{\partial^2 T}{\partial x^2} - \nu(T - T_\infty)$$

$$\frac{T_i^{n+1} - T_i^n}{\Delta t} = \alpha \left( \frac{T_{i+1}^{n+1} - 2T_i^{n+1} + T_{i-1}^{n+1}}{(\Delta x)^2} \right) - \nu(T_i^{n+1} - T_\infty)$$

$$\text{Let F} = \alpha \frac{\Delta t}{(\Delta x)^2}$$

$$-FT_{i-1}^{n+1} + (1 + 2F + \nu \Delta t)T_i^{n+1} - FT_{i+1}^{n+1} = T_i^n + \nu \Delta t T_\infty$$

$$\begin{bmatrix} (1+2F+\nu\Delta t) & -F & 0 & \cdots \\ -F & (1+2F+\nu\Delta t) & -F & \cdots \\ 0 & -F & (1+2F+\nu\Delta t) & -F \\ \vdots & \vdots & -F & (1+2F+\nu\Delta t) \end{bmatrix} \begin{bmatrix} T_1^{n+1} \\ T_2^{n+1} \\ T_3^{n+1} \\ \vdots \\ T_N^{n+1} \end{bmatrix} = \begin{bmatrix} T_1^n + \nu\Delta t T_\infty + FT_L \\ T_2^n + \nu\Delta t T_\infty \\ T_3^n + \nu\Delta t T_\infty \\ \vdots \\ T_N^n + \nu\Delta t T_\infty + FT_R \end{bmatrix}$$

**3. Boundary Treatment (Dirichlet):**
Now, accounting for the initial Dirichlet boundary conditions, we can incorporate the boundary conditions into the system of equations by adding a vector $\mathbf{b}_{bc}$, which enforces the fixed temperatures at the boundaries $(100°\text{F}, 20°\text{F})$, effectively driving the heat flow toward a linear steady-state where $\mathbf{KT} = \mathbf{0}$. meaning there is no net heat flow and the temperature distribution is stable.

$$(\mathbf{I} - \alpha \Delta t \mathbf{K}')\mathbf{T}^{n+1} = \mathbf{T}^n + \mathbf{b}_{bc}$$

## Analysis and Results of General Model

## Real World Data Comparison

### Air Temperature Affect

To make this equation more realistic by incorporating ambient air influence, the 1D heat conduction is now governed by:

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2} - \nu(T - T_\infty)$$

- $\nu$: Convection Coefficient $(s^{-1})$
- $T$ = Temperature (°C),
- $T_\infty$ = Ambient Temperature (°C)

If we convert this to a similar backward Euler scheme, we can derive the following scheme:

$$(\mathbf{I} - \alpha \Delta t \mathbf{K}' + \nu \Delta t \mathbf{I})\mathbf{T}^{n+1} = \mathbf{T}^n + \nu \Delta t T_\infty \mathbf{I} + \mathbf{b}_{bc}$$

This now makes the iteration scheme a bit more complicated than before, yet all we've actually done is add more constants to both sides of the equation to involve the ambient temperature.

**Using a Real World Dataset**

We are going to use a pretty large dataset of temperature measurements along a rod over time, which was collected by students at the Missouri University of Science and Technology. This dataset also provides extra parameters for different types of metals and a setting for including wind moving the air outside of the rod.

**Important mention: In the main corner models, we had a similar number of space and time points, but this model has way more time points to compensate for barely any space points**

From what we mentioned before, this dataset is unique because the experiement only had 6 points along the rod that measureed the temperateure, but they had 4000 time steps, which compensates for the few spatial points and allows us to have a more accurate model with such a high degree of precision.

### What is this provided code doing?

First off, Magic. Secondly, we were given a list of material properties and respective settings

- Aluminum (Al):
  Density = 2700 $kg/m^3$, Thermal Conductivity = 205 $W/m \cdot ^\circ C$, Specific Heat = 900 $J/kg \cdot ^\circ C$
- Stainless Steel (SS):
  Density = 8000 $kg/m^3$, Thermal Conductivity = 16 $W/m \cdot ^\circ C$, Specific Heat = 500 $J/kg \cdot ^\circ C$
- Copper (Cu):
  Density = 8900 $kg/m^3$, Thermal Conductivity = 385 $W/m \cdot ^\circ C$, Specific Heat = 385 $J/kg \cdot ^\circ C$
- 3D Printed Stainless Steel (ADD):
  Density = 8000 $kg/m^3$, Thermal Conductivity = 16 $W/m \cdot ^\circ C$, Specific Heat = 500 $J/kg \cdot ^\circ C$
  (but with different thermal properties based on printing direction)

Material and environmental settings for the dataset:

- Temp = Bool: Allows for a custom initial temperature instead of starting at 0F
- Wind = Bool: Allows for wind to push and affect the temperature outside of the rod,
- Al = Bool: Indicates the use of aluminum,
- SS = Bool: Indicates the use of stainless steel,
- Cu = Bool: Indicates the use of copper,

3D printed Stainless Steel settings for more nuanced thermal properties:

- ADD = Bool: Indicates the use of 3D printed Stainless Steel
- XN = Bool: Printing the rod along the X-negitive direction

- XP = Bool: Printing the rod along the X-positive direction
- YN = Bool: Printing the rod along the Y-negitive direction
- YP = Bool: Printing the rod along the Y-positive direction

These setings are a bit confusing at first, but due to the nature of 3D printing, depending on the leading direction of printing, the cooling fan, can affect extrution and the drying progress so the densitiy and thermal properties of the rod are slightly different.

Depending on the options we choose, the code will pull an excel file provided by those students that are about 4000 rows by 8 columns of data. Then the code compiles it together and from the best of my understanding, does the following:

1. **Aquiring Data**: Pulls all the temperature data at the 6 points along the rod over time given our chosen settings
2. **Building the Matrix Model**:Fits all the needed data into a PDE approximating matrix along with the boundary conditions and the initial temperature gradient
3. **Boundary Gradient Approximation**:Iteratively applies a second order backward finite difference scheme best approximate points at and near the boundaries. This solves for $\frac{\partial T}{\partial x}$ at the boundaries to know how much heat enters the rod.
4. **Steady State Solution Approximation**:Then approximates the points near the end of the time range for the steady state solution. This solves for $\nu$ (convection coefficient) and an initial guess for $k$ (thermal conductivity) for the material and settings we chose
5. **Transient Modeling**:Inputs the predicted temperatures into a 100 term fourier series to build a nice continuous function from distrete points. This solves for a very good approx for $\alpha$ (speed of the heat) and $k$ (conductivity).
6. **Optimization**:Compares the predicted temperatures with the actual temperatures using a least squares error metric to see how well the model is doing. This solves for the error of the model and how well it fits the real world data.
7. **Gathering Parameters**: This Magic Code found a great approximation for the following values: $\nu$, $k$, and $\alpha$ and $\frac{\partial T}{\partial y}$ at the boundaries, now we can substitude these values back into the PDE
8. **Visual Comparison**:Then plots the predicted and actual temperature values over time to visually compare the model's performance against the real-world data.

### Summary Please its too much

$$(\mathbf{I} - \alpha \Delta t \mathbf{K}')\mathbf{T}^{n+1} = \mathbf{T}^n + \mathbf{b}_{bc}$$

$$\Downarrow \qquad \Downarrow \qquad \Downarrow$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{100} a_n \cos(nx) + b_n \sin(nx)$$

$$\Downarrow \qquad \Downarrow \qquad \Downarrow$$

$$S = \sum_{i=1}^{n} (y_i - f(x_i))^2$$

**What does the code tell us?**

Table 2: Thermal Parameter Comparison: Bulk Copper vs. 3D-Printed Steel (XN)

| **Property** | **Copper** | **3D Printed** | **Physical Interpretation** |
|---|---|---|---|
| Conductivity ($k$) | 384.72 W/m·°C | 14.40 W/m·°C | 3DP is ≈ 26x more resistive to heat flow than Cu. |
| Diffusivity ($\alpha$) | 0.0738 $m^2$/s | 0.0213 $m^2$/s | Cu reaches steady state ≈ 3.5x faster than 3DP. |
| Convection ($h$) | 165.41 $W/m^2 \cdot ^\circ C$ | 128.70 $W/m^2 \cdot ^\circ C$ | Variations due to ambient lab/wind conditions. |
| Avg. Std. Error | 0.95 | 0.73 | Low error validates 1D FEA model robustness. |

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2} - \nu(T - T_\infty) \Longrightarrow \rho c_p \frac{\partial T}{\partial t} = k \frac{\partial^2 T}{\partial x^2} - \nu(T - T_\infty)$$

VALUES FOR Xn1-5, redo for Cu1-5 on Windows

- $\rho \approx 7828.79$
- $c_p \approx 502$
- $k \approx 11.313$
- $\nu \approx 0.096$
- $\alpha \approx 0.224$
- with $T_\infty = 20.8$

$$\frac{\partial T}{\partial t} = 0.022387 \frac{\partial^2 T}{\partial x^2} - 0.096192(T - 20.8)$$

**How accurate is our model?**

Quite accurate! The images below show that both plots look nearly identical. Honestly surprised we could take 4000 points per thermocouple and get such a good fit with only 6 spatial points. The code also outputs a highly detailed chart of all the obtained values, with standard error, the unknown parameter estimation errors, and more.

**Important Mention: Show some example graphs and talk about it and computation time**

## Analysis and Results of Real World Model

## Discussion

## Conclusions

## Reflection (1 Page)

- Include a short reflection addressing:
- What feedback did you receive on your poster?
- What did you improve in the presentation and paper?
- What mathematical aspects did you deepen?
- What did you learn about communicating applied math?

### Audio Feedbackn from Poster

1. well executed focus and application
2. stronger applied poster in the class
3. bit of visual clutter (images too big) and more story
4. clear images but print whole poster
5. great font sizes to see from a distance
6. main thing is add more detailed images and less frozen either and hard to itentify final result since they look too similar
7. great analysis of heat equation and details behind them and how the method works
8. really like i took olivers suggestion for real world data thats a strong element of the poster and great contant i can keep leaning on
9. main take away is clear but modify the first parts of story and last part is great
10. what did we learn feedback, a box for the take away, not obvious for conclusion for reader but great understanding
11. need to show off the references more in poster but pretty good 98

## References (1 Page)


### General Links for Sources about FEA

- **"Why is a Triangular Element Stiffer?"** *EnterFEA.*
  https://enterfea.com/why-is-a-triangular-element-stiffer/
- **"The Importance of Mesh Convergence."** *NAFEMS Knowledge Base.*
  https://www.nafems.org/publications/knowledge-base/the-importance-of-mesh-convergence-par
- **"What is Finite Element Analysis and How Does It Really Work?"** *Fidelis FEA.*
  https://www.fidelisfea.com/post/what-is-finite-element-analysis-and-how-does-it-really-wo
- **"What Is FEA — Finite Element Analysis?"** *SimScale Documentation.*
  https://www.simscale.com/docs/simwiki/fea-finite-element-analysis/what-is-fea-finite-elem
- **"What Is Finite Element Analysis?"** *MathWorks Discovery.*
  https://www.mathworks.com/discovery/finite-element-analysis.html
- **"Solving Partial Differential Equations with Finite Elements."** *Wolfram Reference.*
  https://reference.wolfram.com/language/FEMDocumentation/tutorial/SolvingPDEwithFEM.html

### Links for Thermal Property Values Estimator Infomation and Coding

- **"Material thermal properties estimation via a one-dimensional transient convection model."** *Applied Thermal Engineering*, ScienceDirect.
  https://doi.org/10.1016/j.applthermaleng.2020.116230
- **"Data on the Validation to Determine the Material Thermal Properties Estimation."** *Data in Brief*, ScienceDirect.
  https://doi.org/10.1016/j.dib.2021.107577
- **"Python code for 1D Multiple Parameter Estimator."** *Zenodo Repository.*
  https://zenodo.org/records/5683312
- **"oneDkhEstimator Required Dataset."** *Mendeley Data.*
  https://data.mendeley.com/datasets/sbcf36976g/1
- **"Gemini 3 Flash."** *Google AI.* https://gemini.google.com.
  Used for generating 3D tube diagram code, used for understanding the provided magic code.

### Links to Presentation and Poster

- **Presentation Slides.** *Google Slides.*
  https://docs.google.com/presentation/d/1VPHus1hrQ4LEmZuCRAmAq4DViZxpAG3NeTNOqDCx5iA/edit?usp=sharing

- **Project Poster.** *Google Slides.* https://docs.google.com/presentation/d/12PqFTOBQ9jGK5aHqCC0VSaQxrgSarbEYJ4f69zTSPOY/edit?slide=id.p#slide=id.p

Table 3: Thermal Parameter Comparison: Bulk Copper vs. 3D-Printed Steel (XN)

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| Avg. Std. Error | 0.95 |

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